

MD. REZWAN AHMED RATUL

Linux Tinkerer | DevOps Learner
Competitive Programmer

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Aspiring DevOps Engineer with hands-on experience in Linux system administration, Docker containerization, Proxmox virtualization, and network infrastructure management. Passionate about automation, self-hosting, and Infrastructure as Code, with a strong foundation in competitive programming. Seeking internship or entry-level opportunities in DevOps and cloud infrastructure.

TECHNICAL SKILLS

Linux & Operating Systems : NixOS, Arch, Ubuntu, Debian, Kali, Fedora, Windows Server
Containers & Virtualization : Docker, Docker Compose, Proxmox VE (Hypervisor), KVM, LXC, Kubernetes / K3s (Basics)
Infrastructure as Code & CI/CD : Nix Flakes, Declarative Systems, GitHub Actions, Git, GitOps Workflows, Gitea
Networking, Security & VPN : OPNsense, OpenWrt Firewall, WireGuard VPN, VLAN Segmentation, DNS/DHCP, MWAN3 Load Balancing, SSH, SOCKS5
Web & Reverse Proxies : Traefik, Nginx, Nginx Proxy Manager, Cloudflare Pages, REST APIs, TLS/SSL Certificates
Programming & Web Development : Python, Bash, JavaScript, React, C/C++, Java, PostgreSQL, MongoDB, MySQL

DEVOPS & INFRASTRUCTURE PROJECTS

Enterprise Homelab Infrastructure | *Proxmox VE, Docker, OPNsense, WireGuard, Traefik* Jan. 2024 – Present

- Set up and maintain a private cloud on **Proxmox VE** running **5 VMs**, **12 LXC containers**, and **20+ Docker** services managed via **Docker Compose**, sustaining **97.5%** uptime over 12 months.
- Secured the network with an **OPNsense firewall**, **4 VLANs**, internal **DNS** for 30+ host records, and TLS-encrypted reverse proxy routing via **Traefik** and **Nginx Proxy Manager**.
- Configured an encrypted **WireGuard VPN** tunnel on OpenWrt with **MWAN3 failover** across **2 ISPs**, reducing downtime by 90%; scheduled nightly backups to **Gitea** and offsite storage via cron jobs.

Declarative Systems Architecture | *Nix, NixOS, Flakes, Systemd, Git* | [GitHub](#) Jan. 2025 – Present

- Built a reproducible Linux setup with **NixOS** and **Nix Flakes**, managing **200+ packages** and **3,000+ lines** of configuration across **2 machines** (workstation + test server).
- Tracked **100+ dependencies**, dotfiles, user profiles, and **5 custom Systemd** services in a version-controlled **Git** repository, eliminating configuration drift and enabling single-command rollbacks.
- Configured **Home Manager** alongside **Nix Flakes** to declaratively manage terminal tools, user profiles, and desktop settings — all tracked alongside the system config.

SockPit – Infrastructure Proxy Manager | *Python, Windows Server, REST API* | [GitHub](#) July 2026 – Present

- Developed a proxy management tool in **Python (2,500+ lines)** to provision, monitor, and control **SOCKS5** connections on **4 Windows Server** nodes over REST APIs.
- Added user authentication for **50+ managed ports**, dynamic port allocation, and IP whitelisting; configured **Systemd** and Windows services to auto-restart failed processes within **5 seconds**.
- Set up a **3-stage CI** pipeline in **GitHub Actions** — code analysis, **25+ unit tests**, and cross-platform binary builds — reducing manual release work by **80%**.

PlayFlix – Streaming App & Edge CDN | *Cloudflare Pages, REST API* | [Demo](#) · [GitHub](#) July 2026 – Present

- Deployed a serverless web streaming app with access to **1M+ TMDB catalog entries** on **Cloudflare Pages** with external streaming servers, achieving sub-100ms response times worldwide.
- Automated CD pipelines to run build checks and deploy on every code push, reducing deploy time from **10 minutes to under 45 seconds**.
- Added client-side caching for TMDB API data, cutting external API calls by **40%** and saving **1,200+ requests/day**.

Smart Torrent Router Extension | *JavaScript, WebExtensions, Web API* | [Add-on](#) · [GitHub](#) Feb. 2025 – Apr. 2025

- Published an open-source browser extension on **Mozilla Add-ons (500+ lines JavaScript)** using the WebExtensions API to send torrents to a download client in one click.

- Engineered **3 regex-based** interceptors to capture magnet URIs and forward them securely via HTTP POST to qBittorrent Web APIs with token-based session authentication, handling **10+ link formats**.

Decentralized Blockchain Voting System | *JavaScript, Ethereum, Ganache* | [GitHub](#) Jan. 2026 – April 2026

- Built a voting application on **Ethereum (4 Solidity smart contracts)** where ballots are stored on-chain and verified cryptographically, preventing tampering.
- Validated **50+ test transactions**, gas efficiency (reduced costs by **30%**), and voter privacy on simulated blockchain testnets with **Ganache**.

EDUCATION

Khwaja Yunus Ali University

Sirajganj, Bangladesh

BSc in Computer Science & Engineering (Third-Year Student) | CGPA: 3.73 / 4.00 Expected Graduation: July 2028

- **Relevant Coursework:** Computer Networks, Database Systems, Operating Systems, Data Structures, Algorithms, Software Engineering.

COMPETITIVE PROGRAMMING & ACHIEVEMENTS

- **ICPC Asia West Regionals 2025:** Qualified and solved 2 problems in the premier international collegiate competitive programming contest.
- **Bangladesh ICT Olympiad 2025 (DUET):** Ranked **11th nationally** among university participants across Bangladesh.
- **University Programming Contests (KYAU):** Awarded **1st Runner-up (2025)** and achieved multiple 1st-place campus contest victories.
- **Algorithmic Problem Solving:** Solved 140+ problems on [Codeforces](#) (current rating: 888, max: 1042, 35+ rated rounds); active on [Beecrowd](#) and [CodeChef](#). Focused on time/space complexity, data structures, and debugging under contest time limits.